

[Boron: Fact Sheet for Consumers \(original version\)](#)

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What is boron and what does it do?

Boron is a mineral in many foods. Scientists aren't certain what role, if any, boron has in the body, so they don't consider boron to be an essential nutrient.

How much boron do I need?

Experts haven't set a recommended amount of boron.

What foods provide boron?

Many foods, especially from plants, contain boron. These foods include the following:

- Fruits and fruit juices, such as raisins, peaches, and prune and grape juices
- Avocados and potatoes
- Legumes, such as peanuts, beans, and green peas
- Coffee, milk, cider, wine, and beer

What kinds of boron dietary supplements are available?

Boron is present in dietary supplements in a variety of forms. These forms include boron aspartate, boron citrate, boron gluconate, boron glycinate, and calcium fructoborate. Scientists don't know if one form of boron is better than others.

How much boron do people get?

Most adults in the United States get about 1 milligram (mg) of boron a day from food. People who eat more plant foods tend to get more boron than people who eat fewer plant foods.

What happens if I don't get enough boron?

Scientists aren't sure if low intakes of boron cause any harm. A few studies suggest that people who consume low amounts of boron might have more trouble staying mentally alert and focused. Getting low amounts of boron might also lower bone strength.

What are some effects of boron on health?

Scientists are studying boron to understand whether it affects health. Here are some examples of what this research has shown.

Osteoarthritis and bone health

Some studies have tried to find out if boron reduces osteoarthritis symptoms, perhaps by lowering inflammation. Other studies have tried to find out if boron helps keep bones healthy. More research is needed to understand the effects of boron on these conditions.

Cancer

Researchers are studying whether boron could help lower the risk of cancer. More studies are needed to learn whether it has any effect.

Can boron be harmful?

Boron in foods and beverages isn't harmful. But boron can cause harm if a person accidentally swallows cleaning products or pesticides that contain certain forms of boron, such as borax (sodium borate) or boric acid.

The symptoms of too much boron include nausea, vomiting, diarrhea, rashes, headaches, and convulsions. Very high amounts of boron can cause death.

The daily upper limits for boron include intakes from all sources—food, beverages, and supplements—and are listed below in milligrams (mg).

Ages	Upper Limit
Birth to 6 months	Not established
Infants 7–12 months	Not established
Children 1–3 years	3 mg
Children 4–8 years	6 mg
Children 9–13 years	11 mg
Teens 14–18 years	17 mg
Adults	20 mg
Pregnant and breastfeeding teens	17 mg
Pregnant and breastfeeding women	20 mg

Does boron interact with medications or other dietary supplements?

Boron is not known to interact or interfere with any medicines or dietary supplements.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if these dietary supplements might interact with your medicines. They can also explain whether these medicines might interfere with how your body absorbs or uses boron or other nutrients.

Healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about boron?

- For more information on boron and food sources of boron
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Boron](#)
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

infant

A child younger than 12 months old.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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